

LinkedIn open sources a tool that formats Big Data for TensorFlow

LinkedIn has open sourced a tool it developed to convert Apache Spark-based Big Data into a format that can be readily consumed by TensorFlow, the leading deep learning library.

The tool, called Avro2TF, aims to reduce the complexity of data processing and improve the velocity of advanced modelling.

With this technology, developers can improve productivity by focusing on building machine learning (ML) models rather than converting data, said sources at LinkedIn, which is on a mission to democratise machine learning.

“Based on the feedback from our users on the LinkedIn ML vertical teams, we needed a scalable solution focused on scalable data conversion. More specifically, we needed a solution that converted our LinkedIn data types (e.g., sparse vector, dense vector, etc) into a deep learning format (i.e., tensors),” the LinkedIn engineering team explained in a blog post.

The team members noted that like LinkedIn, many other companies have vast amounts of ML data in a similar sparse vector format, and the tensor format is still relatively new to many of them. They believe that Avro2TF can help address these challenges.

“Avro2TF bridges the gap and presents an elegant solution for ML engineers, freeing them up to focus on different deep learning algorithms,” the team members stated.

Avro2TF provides a simple configuration for modellers to obtain tensors from existing training data. Tensor data itself is not self-contained. So, to be loaded to TensorFlow, it is required to carry metadata, and Avro2TF fills this gap by providing a distributed metadata collection job.

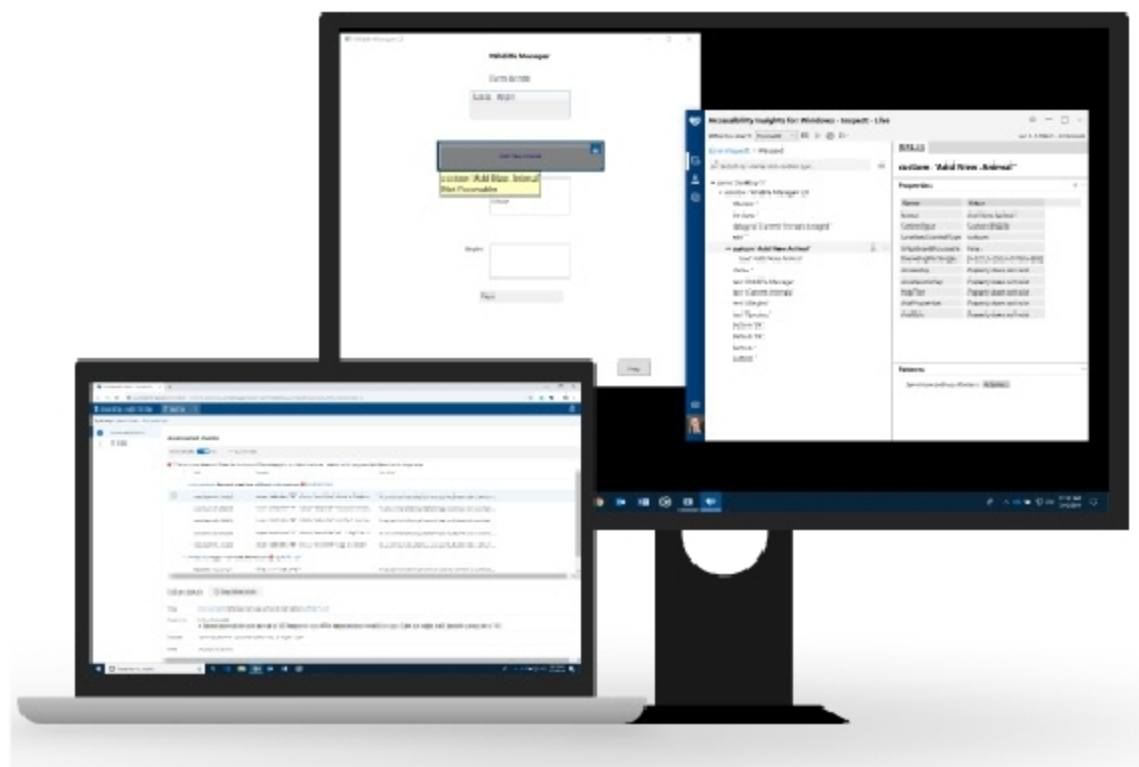
Inside LinkedIn, Avro2TF is an integral part of a system called TensorFlowIn that helps users easily feed data into the TensorFlow modelling process.

Microsoft's Accessibility Insights apps now available as open source code

Microsoft reiterated that it is on a journey to design, build and launch more accessible products to foster inclusion by announcing the open sourcing of two of its Accessibility Insights applications.

The two free tools, Accessibility Insights for Windows and Accessibility Insights for Web, help developers find and fix common accessibility issues early in the software development cycle.

Built on open source Axe technology from Deque Systems, Accessibility Insights can be used as standalone tools or can be integrated through their rules engines into a developer's build process, Keith Ballinger, a developer services general manager with Microsoft, wrote in a post on the company's Open Source Blog.



Ballinger said, “We are driven by the promise of more accessible products for more people. That’s why we’re releasing Accessibility Insights to the open source and accessibility communities – it’s all of ours now, and together we’ll continue to make it a better tool and build a more accessible future.”

Ballinger also announced that Deque Systems is providing GitHub issue-filing for Accessibility Insights for Web and colour contrast detection heuristics for Accessibility Insights for Windows. Microsoft contributed its Windows rules engine to the Axe Accessibility Project so all major platforms could be covered.

The automated checks in Accessibility Insights can identify common accessibility issues in seconds. This enables the developer to find and fix any issues detected through automated checks before proceeding to the remaining tests, which require human judgement. The human-powered tests are supported by assistance from the tools, which makes these tests easier, faster, and more reliable.

“By focusing on easy-to-use tooling, we hope to help make accessibility more accessible,” Ballinger wrote. He added, “Our aim with Accessibility Insights is not to reinvent the wheel, but to modernise existing technology and optimise it for use in developer workflows. It’s a piece of a much larger inclusion effort – one that involves inclusive hiring, Trusted Tester – all the tools in the toolbox!”

Accessibility Insights for Web has been open sourced on GitHub.